



Province of the
EASTERN CAPE
EDUCATION

Iphondo leMpona Kapa: Isebe leMfundo
Provinsie van die Oos Kaap: Departement van Onderwys
Porafensie Ya Kapa Botjhabela: Lefapha la Thuto

**NATIONAL
SENIOR CERTIFICATE/
*NASIONALE
SENIOR SERTIFIKAAT***

GRADE/*GRAAD* 11

NOVEMBER 2025

**TECHNICAL MATHEMATICS P2/*TEGNIIESE WISKUNDE V2*
MARKING GUIDELINE/*NASIENRIGLYN***

MARKS/*PUNTE*: 150

This marking guideline consists of 20 pages./
Hierdie nasienriglyn bestaan uit 20 bladsye.

NOTE:

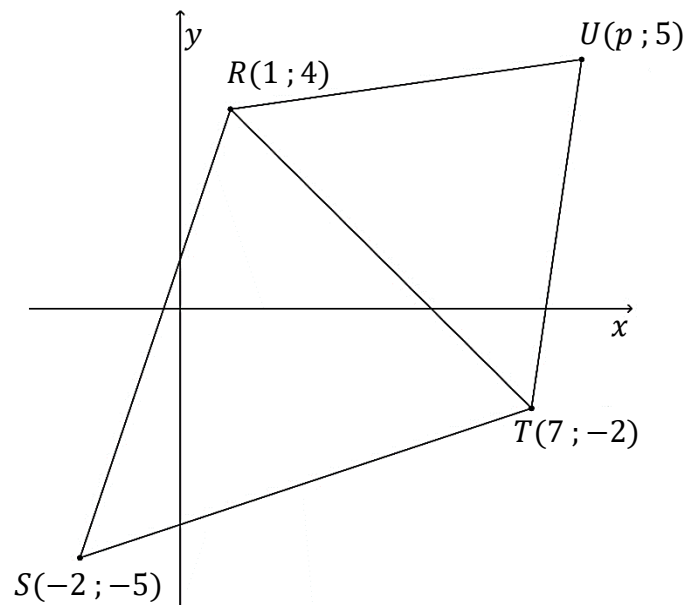
- Continuous accuracy (CA) applies only where indicated in this marking guideline.
- Assuming values/answers in order to solve a problem is unacceptable.

LET WEL:

- *Volgehoue akkuraatheid (CA) is slegs van toepassing soos aangedui in hierdie nasienriglyn.*
- *Aanvaarding van waardes/antwoorde om 'n probleem op te los, is onaanvaarbaar.*

MARKING CODES/NASIENKODES	
M	Method/Metode
A	Accuracy/Akkuraatheid
AO	Answer only/Slegs antwoord
CA	Consistent accuracy/Deurlopende akkuraatheid
F	Formula/Formule
I	Identity/Identiteit
R	Rounding/Afronding
S	Simplification/Vereenvoudiging
ST	Statement/Bewering
RE	Reason/Rede
ST RE	Statement and correct reason/Bewering en korrekte rede
SF	Substituting correctly in the correct formula/Korrekte vervanging in die korrekte formule
NPU	No penalty for omitting units/Geen penalisering vir eenhede uitgelaat
NPR	No penalty for rounding/Geen penalisering vir afronding

QUESTION/VRAAG 1

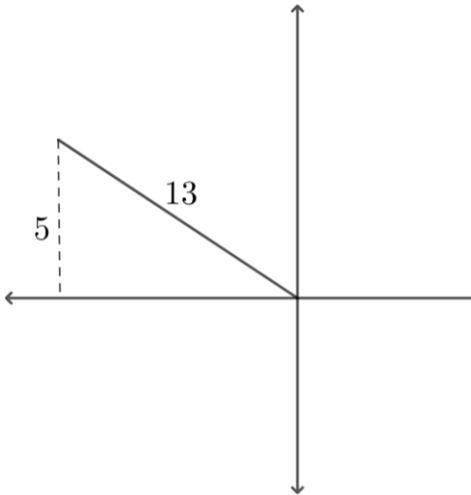


1.1	$m_{RT} = \frac{y_2 - y_1}{x_2 - x_1}$ $= \frac{-2 - 4}{7 - 1}$ $= -1$	✓ S A ✓ m_{RT} CA	(2)
1.2	$Q\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$ $= Q\left(\frac{1 + 7}{2}; \frac{4 + (-2)}{2}\right)$ $= M(4; 1)$	✓ x-value/ x-waarde A ✓ y-value/ y-waarde A	(2)

1.3	$RT = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $RT = \sqrt{(7 - 1)^2 + (-2 - 4)^2}$ $RT = \sqrt{6^2 + (-6)^2}$ $RT = 6\sqrt{2} \text{ units / eenhede}$	✓ SF CA ✓ S CA ✓ RT CA	(3)
1.4	$m_{SQ} = \frac{1 - (-5)}{4 - (-2)}$ $= 1$ $m_{RT} \times m_{SQ} = -1 \times 1$ $= -1$	✓ m_{SQ} CA ✓ Su CA ✓ Prod CA	(3)
1.5	$m = m_{RT} = -1$ $y - y_1 = m(x - x_1)$ $y - (-5) = -1(x - (-2))$ $y = -x - 2 - 5$ $y = -x - 7$	✓ $m = -1$ CA ✓ SF A ✓ S CA ✓ Answer/ Antwoord CA	(4)
1.6	$\sqrt{(p - (-2))^2 + (5 - (-5))^2} = 10\sqrt{2}$ $\sqrt{(p + 2)^2 + (10)^2} = 10\sqrt{2}$ $(p + 2)^2 + (10)^2 = (10\sqrt{2})^2$ $(p + 2)^2 = 200 - 100$ $p + 2 = \pm\sqrt{100}$ $p = -2 \pm 10$ $p = 8 \text{ and/en } p \neq -12$	✓ SF A ✓ Squaring/ Kwadratuur CA ✓ Transpose/ Transponeer CA ✓ S CA ✓ $p = 8$ CA	(5)

1.7	$SQ = \sqrt{(4+2)^2 + (1+5)^2}$ $SQ = 6\sqrt{2}$ $RQ = 3\sqrt{2}$ $QT = 3\sqrt{2}$ $\text{Area/oppervlakte } \triangle SQR = \frac{1}{2} RQ \times SQ$ $= \frac{1}{2} \times 3\sqrt{2} \times 6\sqrt{2}$ $= 18 \text{ units}^2 / \text{eenhede}^2$ $\text{Area/oppervlakte } \triangle SQT = \frac{1}{2} QT \times SQ$ $= \frac{1}{2} \times 3\sqrt{2} \times 6\sqrt{2}$ $= 18 \text{ units}^2 / \text{eenhede}^2$ $\text{Area/oppervlakte } \triangle SQR = \text{Area/oppervlakte } \triangle SQT$ $\therefore SQ \text{ divides } \triangle SRT \text{ into two equal parts/SQ verdeel } \triangle SRT \text{ in twee gelykedele}$	✓SF A ✓ SQ CA ✓ SF A ✓ Area/Oppervlakte $\triangle$ SQR CA ✓ SF CA ✓ Conclusion/Slot CA	(6)
			[25]

QUESTION/VRAAG 2

2.1.1	$\sin(207,21^\circ - 22,32^\circ)$ $= -0,09$	✓ Subst/Vervanging ✓ Answer/Antwoord	(2)
2.1.2	$= \frac{13}{\left(\frac{1}{\tan(2 \times 22,31^\circ + 207,21^\circ)} \right)^2}$ $= 120,69$	✓ Subst/Vervanging A ✓ $\frac{1}{\tan}$ A ✓ Answer/Antwoord CA	(3)
2.2.1	 $x^2 + y^2 = r^2$ $x^2 + 5^2 = 13^2$ $x^2 = 169 - 25$ $x = \pm\sqrt{144}$ $x = -12$ $\cos \theta = \frac{-12}{13}$	✓ Diagram A ✓ SF A ✓ $x = -12$ CA ✓ $\cos \theta$ CA	(4)
2.2.2	$\tan \theta = -\frac{5}{12}$	✓ S A	(1)

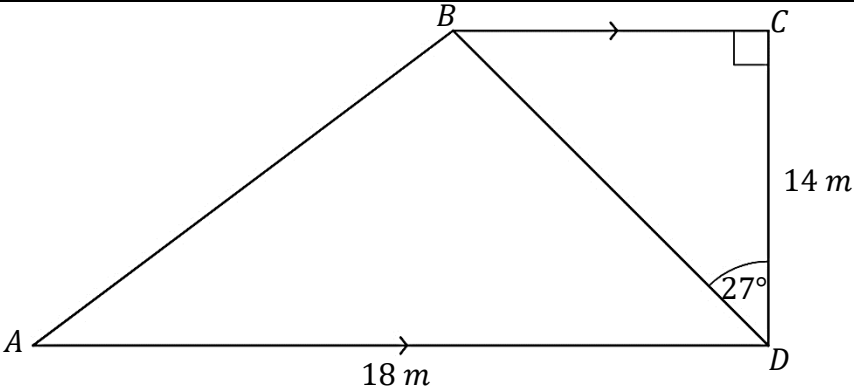
2.2.3	$\cos ec^2 + \sec^2 = \left(\frac{13}{5}\right)^2 + \left(-\frac{13}{12}\right)^2$ $= \frac{28561}{3600}$ $= 7,93$	$\checkmark \left(\frac{13}{5}\right)$ A $\checkmark \left(-\frac{13}{12}\right)$ CA $\checkmark$ Answer/Antwoord CA NPR	(3)
2.3	$5 \cot \beta + 7 = 13$ $5 \cot \beta = 6$ $\cot \beta = \frac{6}{5}$ $\therefore \tan \beta = \frac{5}{6}$ $\beta = \tan^{-1}\left(\frac{5}{6}\right)$ Ref $\angle = 39,81^\circ$ 1 st Quadrant/Kwadrant $\beta = 39,81^\circ$ 3 rd Quadrant/Kwadrant $\beta = 180^\circ + 39,81^\circ$ $\beta = 219,81^\circ$	$\checkmark$ S A $\checkmark \tan^{-1}\left(\frac{5}{6}\right)$ A $\checkmark$ Ref $\angle$ CA $\checkmark \beta = 39,81^\circ$ CA $\checkmark \beta = 219,81^\circ$ A	(5)
			[18]

QUESTION/VRAAG 3

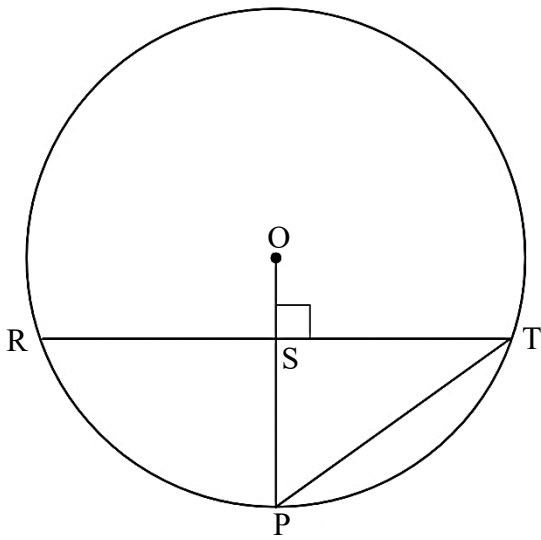
3.1	$\frac{\cos(360^\circ - x) \cdot \sin(180^\circ + x) \cdot \tan(180^\circ + x)}{\cos^2(180^\circ + x) \cdot \cos \pi}$ $= \frac{\cos x \cdot \sin x \cdot \tan x}{(-\cos x)^2 \cdot (-1)}$ $= \frac{\cos x \cdot (-\sin x) \cdot \frac{\sin x}{\cos x}}{(-\cos x)^2 \cdot (-1)}$ $= \frac{\sin^2(x)}{\cos^2(x)}$ $= \tan^2 x$ OR/OF $= \sec^2 x - 1$	$\checkmark \cos x$ A $\checkmark \sin x$ A $\checkmark \tan x$ A $\checkmark (-\cos x)^2$ A $\checkmark \frac{\sin x}{\cos x}$ A $\checkmark (-1)$ A $\checkmark \text{Answer/Antwoord}$ CA	
3.2	$\frac{\cos x - \cos x \cdot \sin x}{\sin x - 1 + \cos^2 x} = \cot x$ $LHS = \frac{\cos x(1 - \sin x)}{\sin x - 1 + 1 - \sin^2 x}$ $= \frac{\cos x(1 - \sin x)}{\sin x(1 - \sin x)}$ $= \frac{\cos x}{\sin x}$ $= \cot x$ $= RHS$	$\checkmark (1 - \sin x)$ A $\checkmark 1 - \sin^2 x$ A $\checkmark S$ A $\checkmark \frac{\cos x}{\sin x}$ CA	(7)
			(4)
			[11]

QUESTION/VRAAG 4				
4.1			<i>f:</i> ✓ shape/vorm ✓ start and end point/ <i>begin- en eindpunt</i> ✓ Asymptotes/<i>asimptote</i> <i>g:</i> ✓ intercepts/<i>afsnitte</i> ✓ shape/vorm ✓ start and end point/ <i>begin- en eindpunt</i> ✓ turning points/ <i>draaipunte</i>	(7)
4.2	$x = 90^\circ$ $x = 270^\circ$	✓ $x = 90^\circ$ ✓ $x = 270^\circ$	A A	(2)
4.3	$Per_g = 360^\circ$	✓ Answer/ <i>Antwoord</i>	A	(1)
4.4	$0^\circ < x < 90^\circ$ $150^\circ < x < 180^\circ$ $270^\circ < x < 330^\circ$	✓ $0^\circ < x < 90^\circ$ ✓ $150^\circ < x < 180^\circ$ ✓ $120^\circ < x < 330^\circ$	A A	(3)
				[13]

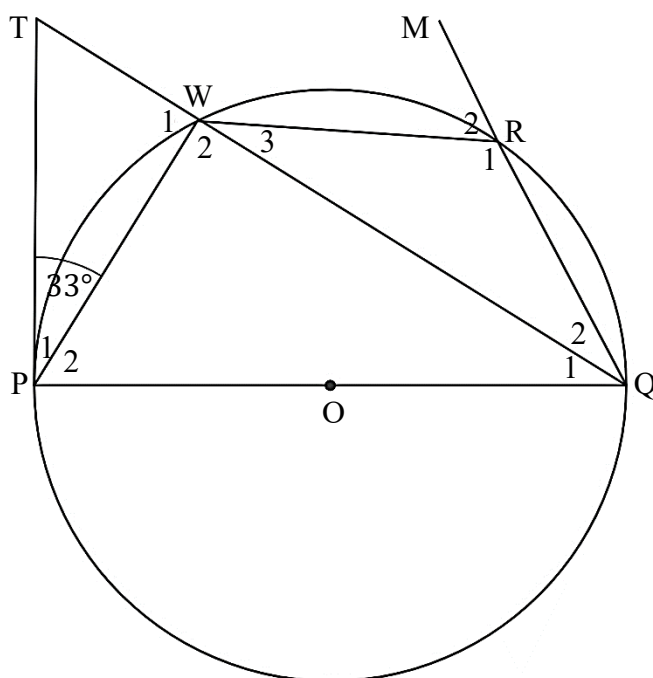
QUESTION/VRAAG 5

			
5.1	$\cos 27^\circ = \frac{CD}{BD}$ $BD = \frac{14}{\cos 27^\circ}$ $BD = 15,71m$ OR $\sin 27^\circ = \frac{CD}{BD}$ $BD = \frac{14}{\sin 63^\circ}$ $BD = 15,71m$	$\checkmark \cos 27^\circ = \frac{CD}{BD}$ A $\checkmark BD$ A OR/OF $\checkmark \sin 27^\circ = \frac{CD}{BD}$ A $\checkmark BD$ A	(2)
5.2	$\hat{A}DB = 63^\circ$ $\hat{D} + \hat{C} = 180^\circ$ $co-int \angle s, BC \parallel AD$ $AB^2 = BD^2 + AD^2 - 2 \cdot BD \cdot AD \cdot \cos \hat{D}$ $AB^2 = 18^2 + (15,71)^2 - (2 \times 18 \times 15,71 \times \cos 63^\circ)$ $SG = \sqrt{314,0452}$ $SG = 17,72m$	$\checkmark$ ST/RE A $\checkmark$ Cosine Rule/ Kosinusreël A $\checkmark$ SF CA $\checkmark$ S CA $\checkmark$ Answer/Antwoord CA	(5)

QUESTION/VRAAG 6

				
6.1(i)	Centre	✓ Answ/Ant	A	(1)
6.1(ii)	Bisects	✓ Answ/Ant	A	(1)
6.2.1	$RS = 12cm$ <i>line from cntr bisects the chrd</i> $OP^2 = OS^2 + RS^2$ <i>Pythagoras Thm</i> $OP = \sqrt{12^2 + 5^2}$ $OP = \sqrt{169}$ $OP = 13cm$ $SP = 13 - 5$ $= 8cm$	✓ ST/RE A ✓ ST/RE A ✓ OP CA ✓ SP CA		(4)
6.2.1	$PT^2 = SP^2 + ST^2$ <i>Pythagoras thm</i> $PT = \sqrt{8^2 + 12^2}$ $PT = 4\sqrt{13}$ $PT = 14,42cm$	✓ ST/RE A ✓ S CA ✓ PT CA		(3)
				[9]

QUESTION/VRAAG 7



7.1(i)	Exterior/Buite	✓ Answ/Ant	A	(1)
7.1(ii)	Interior/Binne	✓ Answ/Ant	A	(1)
7.2	Tan $\perp$ Rad	✓ RE	A	
7.3.1	$\hat{Q}_1 = 33^\circ$ Tan Chord / Koord	✓ ST ✓ RE	A	(2)
7.3.2	$\hat{P}_2 = 90^\circ - 33^\circ$ Tan $\perp$ Rad $= 57^\circ$ $\hat{R}_2 = \hat{P}_2$ ext $\angle$ cyclic quad / koordevierhoek $= 57^\circ$	✓ ST ✓ ST	A CA	(2)
7.3.3	$\hat{T} + \hat{P} + \hat{Q}_1 = 180^\circ$ sum $\angle$ s in Δ $\hat{T} = 180^\circ - (90^\circ + 33^\circ)$ $\hat{T} = 57^\circ$	✓ ST/RE ✓ $\hat{T}$	A CA	(2)

7.3.4	$\hat{R}_2 = \hat{Q}_2 + \hat{W}_3$ <i>ext $\angle$ of Δ</i> $\hat{Q}_2 = \hat{W}_3$ <i>$\angle$s opp = sides</i> $\hat{Q}_2 = \frac{57^\circ}{2}$ $\hat{Q}_2 = 28,5^\circ$	✓ ST A ✓ RE A ✓ ST A	(3)
7.4	<i>in ΔTQP</i> $\hat{Q} = 33^\circ$ $P = 90^\circ$ Tan $\perp$ Rad $T = 57^\circ$ 3rd $\angle$ <i>in ΔPQW</i> $\hat{Q} = 33^\circ$ $W = 90^\circ$ $\angle$ in semi-circle / sirkel $P = 57^\circ$ 3rd $\angle$ $\therefore \Delta TQP \parallel \Delta PQW$ AAA	 ✓ ST / RE A ✓ ST A ✓ RE A ✓ RE A	(4)
			[16]

8.4	$\hat{R} = \hat{A}_5$ Tan Chord / Koord $\hat{A}_4 = 90^\circ - 62^\circ$ Tan $\perp$ Rad $= 28^\circ$	✓ST A ✓ RE A ✓ST CA	(3)
8.5	<i>in ΔAMO</i> $\hat{A}MO = 90^\circ$ line from centre / lynvan oppervlak $\perp$ to chord / nakoord $\hat{A}_3 = 90^\circ - 86^\circ$ sum of $\angle$ s in Δ / som van $\angle$ s in Δ $= 4^\circ$	✓ST✓ RE A ✓ST CA	(3)
			[15]

QUESTION/VRAAG 9

9.1.1	$\frac{30}{60} = 0,5 rps$		✓ Answer/Antwoord A	(1)
9.1.2	$\omega = 2\pi n$ $= 2 \times \pi \times 0,5$ $= \pi rad / s$ $\approx 3,14 rad / s$	$\omega = 2\pi n$ $= 2 \times \pi \times 30$ $= 60\pi rad / min$ $\approx 188,50 rad / min$	✓ F A ✓ SF CA ✓ Answer/Antwoord CA	(3)
9.1.3	$v = \pi Dn$ $v = \pi \times \left(\frac{27 \times 2}{100} \right) \times 0,5$ $v = \frac{27\pi}{100} m / s$ $v \approx 0,85 m / s$		✓ F A ✓ SF CA ✓ Conversion/Herleiding A ✓ Answer/Antwoord CA NPR, NPU	(4)
9.2.1	$s = r\theta$ $s = \left(\frac{28}{2} \right) \times \left(135^\circ \times \frac{\pi}{180^\circ} \right)$ $s = \frac{21}{2} \pi$ $s = 32,99 cm$		✓ F A ✓ SF A ✓ s CA NPR, NPU	(3)

9.2.2	$\text{Area of sector} = \frac{1}{2}r^2\theta$ $= \frac{1}{2} \times \left(\frac{28}{2}\right)^2 \times \left(\frac{3\pi}{4}\right)$ $= \frac{147}{2}\pi$ $= 230,91\text{m}^2$ OR/OF $\text{Area of sector/Oppervlakte van sektor} = \frac{rs}{2}$ $= \frac{\left(\frac{28}{2}\right) \times \frac{3\pi}{4}}{2}$ $= \frac{147}{2}\pi$ $= 230,91\text{m}^2$	✓F A ✓SF CA ✓θ A ✓Area CA NPU, NPR AO ONE Mark/ EEN Punt OR/OF ✓F A ✓SF CA ✓radius CA ✓Area/oppervlakte CA NPU, NPR AO ONE Mark/ EEN Punt	(4)
9.3	$4h^2 - 4dh + x^2 = 0$ $4h^2 - 4(2 \times 30)h + \left(\frac{200}{10}\right)^2 = 0$ $4h^2 - 240h + 400 = 0$ $h = \frac{-b \pm \sqrt{b^2 - 4ac}}{4a}$ $h = \frac{-(-240) \pm \sqrt{(-240)^2 - (4 \times 4 \times 400)}}{2 \times 4}$ $h = 58,28 \quad \text{or / of} \quad h = 1,72$ $\therefore h = 1,72$	✓F A ✓$\frac{200}{10}$ A ✓SF A ✓Std form/ CA Std vorm ✓S CA ✓h = 1,72 CA NPR, NPU	(6)
			[21]

QUESTION/VRAAG 10			
10.1	$A_T = a \left(\frac{o_1 + o_8}{2} + o_2 + o_3 + o_4 + o_5 + o_6 + o_7 \right)$ $= \frac{320}{8} \left(\frac{140 + 155}{2} + 147 + 150 + 138 + 166 + 170 + 160 \right)$ $= 40(1078,5)$ $= 43140 \text{ m}^2$	✓ F A ✓ SF A ✓ $a = \frac{320}{8}$ A ✓ S CA ✓ Area CA	(5)
	OR/OF $A_T = a(m_1 + m_2 + m_3 + m_4 + \dots + m_n)$ $= \frac{320}{8} \left(\frac{140 + 147}{2} + \frac{147 + 150}{2} + \frac{150 + 138}{2} + \frac{138 + 166}{2} \right.$ $\left. + \frac{166 + 170}{2} + \frac{170 + 160}{2} + \frac{160 + 155}{2} \right)$ $= 40(1078,5)$ $= 43140 \text{ m}^2$	✓ F A ✓ SF A ✓ $a = \frac{320}{8}$ A ✓ S CA ✓ Area/Oppervlakte CA	(5)
10.2.1	$V = lbh$ $V = 0,35 \times 0,2 \times 0,6$ $V = 0,042 \text{ m}^3$ $V = 0,042 \times 1000$ $V = 42 \text{ l}$	 ✓ SF CA ✓ Conversion/ Herleiding CA ✓ Volume CA	(3)

10.2.2	$A_T = 2(lb + lh + bh)$ $= 2((0,35 \times 0,2) + (0,35 \times 0,6) + (0,2 \times 0,6))$ $= 0,8m^2$ <i>Cost of material for 1 bin</i> $= R15,80 \times 0,8$ $= R120,64$ <i>Cost of material for 15 bins</i> $= R120,64 \times 15$ $= R1809,60$	✓ F A ✓ SF CA ✓ <i>Area/Oppervlakte</i> CA ✓ <i>Cost/Koste</i> CA	(4)
			[12]
		TOTAL/TOTAAL:	150